INCH-POUND

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SUPERSEDING
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# PERFORMANCE SPECIFICATION

## LENS ASSEMBLY, OBJECTIVE
## VIEWER, DRIVER'S NIGHT VISION, AN/VVS-2(V)

This specification is approved for use by US Army Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the Objective Lens Assembly, herein known as objective lens, which is part of Viewer, Driver's Night Vision, AN/VVS-2(V).

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specifications or recommended for additional information or as examples. While every effort has been made to insure the completeness of this list, document users are cautioned that they must meet all specified requirements in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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2.1.1 Reference documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.f).

DRAWINGS
USA COMMUNICATIONS-ELECTRONICS COMMAND

SM-D-771334 - Barrel Assembly, Lens, Objective

(Copies of Specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Description. The objective lens assembly is mounted to and is part of the Viewer, Driver's Night Vision AN/VVS-2(V).

3.2 Construction. The objective lens assembly shall be constructed to assure compliance with the contract configuration baseline and in accordance with SM-D-771334 as applicable.

3.3. First article test (FAT). When specified in the contract (see 6.2.e and 6.2.h) the contractor shall furnish units for FAT inspection and approval (see 4.2).

3.4 Materials. The contractor shall select parts and materials that are fully capable of meeting all operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but not mandatory. Selection criteria of the class, grade or type part shall ensure the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.5 Performance characteristics.

3.5.1 Effective focal length. The objective lens shall have an Effective Focal Length (EFL) of 33.6, ± 0.5 millimeters (mm).

3.5.2 Field of view. The field of view shall be not less than 44 degrees for a 25mm diameter format.



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3.5.3 Linear distortion. The objective lens shall have between 9 percent and 12 percent barrel distortion.

3.5.4 Modulation Transfer Function (MTF) on axis. The MTF shall be not less than the values specified in Table I when tested on axis with a sine wave pattern setup that utilizes an S-20-ER response photomultiplier and a tungsten illumination source operated at 2854K ±200K color temperature. The test shall be repeated with the objective rotated 90 degrees about the optical axis.

TABLE I. Objective lens modulation transfer function.

| Test Frequency (cycles/millimeter) on the Image Intensifier | Modulation Transfer (Percent) |
|---|---|
| 10 | 84 |
| 20 | 70 |
| 30 | 55 |
| 40 | 40 |

3.5.5 T-number. The T-number shall be no greater than 1.25 when tested using a light source operated at 2854K ±200K color temperature and a photo-detector with an S-20ER response.

3.5.6 Veiling glare. The veiling glare (stray light) shall be no greater than 2.5 percent when using a photodetector with an S-20ER response.

3.5.7 Flange focal length. The flange focal length shall be not less than 0.039 inches (0.99lmm) when measured from the outermost flange surface of the 1.5-inch diameter central area of the lens retainer.

3.6 Environmental. Verification of conformance after environmental tests shall be Modulation transfer function (3.5.4), Veiling glare (3.5.6) and Flange focal length (3.5.7).

3.6.1 Temperature Extremes. The objective lens shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to storage at +68°C to -51°C. The objective lens shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to repetitive temperature changes between +23°C and +68°C in 5 minutes and between +23°C and -51°C in 5 minutes. Soak time at each temperature extreme shall be sufficient for the objective lens to attain temperature stabilization. Time for temperature stabilization is defined as the amount of time required for the most internal part of the unit under test to achieve the specified temperature.



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3.6.2 Vibration. The objective lens shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to vibration equivalent to that encountered in tracked vehicles during 5000 miles of operation. Vibration shall be limited to one sine sweep from 5.0 to 500 Hertz (Hz) and back to 5.0 Hz within 15 minutes in the vertical axis only. Vibration shall be performed from 5.0 to 5.5 Hz. at 1-inch displacement, 5.5 to 30 Hz. at 1.5 g's peak, 30 to 50 Hz at 0.33-inch displacement and 50 to 500 Hz. at 4.2 g's peak.

3.6.3 Shock.

3.6.3.1 Basic shock. The objective lens shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to basic shock of 40 g's (see 3.10.5) at 18 milliseconds.

3.6.3.2 Weapon shock. The objective lens shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to simulated weapon firing shock of 100 g's at 1.5 milliseconds one half sine pulse.

3.6.4 Immersion. The objective lens shall not be damaged (see 3.10.1) after being immersed in fresh water to a depth of not less than 3 feet for a period of not less than 30 minutes. The objective lens shall be pre-conditioned by exposure to temperature of +45°C ±3°C for a period of 2 hours prior to being immersed in the fresh water at a temperature of +18°C ±5°C.

3.7 Treatment and painting. Treatment and painting shall be anti-reflective, shall be rust resistant and as specified on applicable drawings unless otherwise specified in the contract or purchase order.

3.8 Marking. Marking shall be in accordance with applicable drawings unless otherwise specified in the contract or purchase order.

3.9 Workmanship. Standards of workmanship shall be such that the objective lens will meet all requirements of this specification and any referenced specifications or drawings.

3.10 Technical interpretations. The following technical interpretations are, when referenced in sections 3 or 4, mandatory for this specification.

3.10.1 Damage. Damage is defined as any breakage, loosening, shift, dents, failure of any finish, hardware, connection, or component, or condensation, fogging or moisture on any internal surface.

3.10.2 Ambient temperature. Ambient temperature is defined as +23°C, ± 10°C.

3.10.3 Environmental temperature. Environmental chamber temperature shall be controlled to within ±5°C of the specified temperature unless otherwise specified.



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3.10.4 Degradation of performance. Any deterioration in quality, or failure to operate with the requirements of this specification.

3.10.5 "g". "g" is defined as an acceleration or deceleration of 32.17 feet per second per second.

4. VERIFICATION

4.1 Classification of inspections. Inspections shall be classified as follows:

a. First article tests (4.2)
b. Conformance inspection (4.3)

4.2 First Article Test (FAT). When specified in the contract or purchase order (see 6.2.h), first article testing shall be performed by the contractor. The quantity of first article test samples shall be as specified in the contract or purchase order(see 6.2.e) and shall be selected from the first production lot.

4.2.1 FAT Tests. Each objective lens in the first production lot shall be subjected to the performance tests of Table II. After completion of this test, objective lens assemblies selected as samples for FAT testing shall be subjected to the tests of Table III which may be performed in any order except that Immersion shall be performed last. Failure of any test shall be cause for rejection of that objective lens and may be cause for rejection of the FAT.

TABLE II. Unit first article testing.

| Inspection | Requirement Paragraph |
|---|---|
| Effective focal length | 3.5.1 |
| Field of view | 3.5.2 |
| Linear distortion | 3.5.3 |
| Modulation transfer function | 3.5.4 |
| T-number | 3.5.5 |
| Veiling glare | 3.5.6 |
| Flange focal length | 3.5.7 |

Table III Sample first article testing

| Inspection | Requirement Paragraph |
|---|---|
| Temperature Extremes | 3.6.1 |
| Vibration | 3.6.2 |
| Shock | 3.6.3 |
| Immersion | 3.6.4 |

4.2.1 Disposition of FAT samples. Disposition of FAT samples shall be as specified in the contract or purchase order (see 6.2.e).

4.3 Conformance inspection. Requirements for conformance inspection shall be as specified in the contract or purchase order (see 6.2.g). Unless otherwise specified, all tests shall be performed at ambient temperature (see 3.10.2).



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## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2.c). When packaging is to be performed by DoD personnel, these personnel need to contract the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The objective lens is used with the AN/VVS-2 (V) for driver viewing during periods of darkness.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Necessary actions by contractor in the event of a lot failure (see 4.2.1).
c. Packaging requirements.
d. Sampling plans for lot testing.
e. Quantity and schedule for FAT for all line items and disposition of FAT samples.
f. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.1.1).
g. Statement identifying that the contractor must submit his proposal for conformance inspection to include specific requirements to be verified and the sampling plan to be used to provide the desired confidence level.
h. Statement specifying that FAT testing is, or is not, required.

6.3 Definitions. See 3.10.



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6.4 Subject term keyword listing. None

Effective focal length
Field of view
Linear distortion
Modulation transfer function
T-number
Veiling glare

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

**INSTRUCTIONS**

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| **I RECOMMEND A CHANGE:** | **1. DOCUMENT NUMBER**<br>MIL-PRF-49148D (CR) | **2. DOCUMENT DATE (YYMMDD)**<br>990304 |
|---|---|---|

**3. DOCUMENT TITLE**
Lens Assembly, Objective Viewer, Driver's Night Vision, AN/VVS-2 (V)

**4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)**

**5. REASON FOR RECOMMENDATION**

| **6. SUBMITTER** | | |
|---|---|---|
| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED<br>(YYMMDD) |

| **8. PREPARING ACTIVITY** | |
|---|---|
| a. NAME<br>US Army Communications-Electronics Command | b. TELEPHONE (Include Area Code)<br>(1) Commercial (732) 532-9129<br>(2) AUTOVON 992-9129 |
| c. ADDRESS (Include Zip Code)<br>ATTN: AMSEL-LC-LEO-E-EP<br>Fort Monmouth, NJ 07703-5023 | **IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:**<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |



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